Cardiovascular Disease Prediction: A Machine Learning Approach

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Abstract

Cardiovascular diseases remain a leading cause of mortality worldwide, making early risk identification a critical clinical priority. This study develops a complete machine learning pipeline to predict the presence of heart disease using clinical patient data from the UCI repository (n=920). Four classification algorithms were implemented and compared following data preprocessing, feature engineering and hyperparameter optimization. All models demonstrated strong and comparable predictive performance, with accuracy ranging from 84.6% to 85.2% across all classifiers. Random Forest achieved the highest sensitivity (Recall: 93.0%) and strongest overall reliability (MCC: 0.704), while SVM led on discriminative ability (AUC-ROC: 0.913); Logistic Regression and Gaussian Naive Bayes matched Random Forest's accuracy (85.2%) with marginally superior Macro F1 scores (0.849 and 0.850 respectively). Key predictors identified included cholesterol, maximum heart rate, age, ST depression and exercise-induced angina. The integrative analysis of clinical features and algorithmic performance establishes Random Forest as the optimal model for cardiovascular risk prediction, owing to its superior sensitivity and MCC — critical priorities in clinical screening contexts where minimizing false negatives is paramount.

Keywords: Cardiovascular disease, machine learning, clinical prediction, risk factors, binary classification.

1. Introduction

1.1 Clinical relevance of cardiovascular disease

Cardiovascular diseases (CVDs) represent the leading cause of death worldwide, accounting for approximately 19.8 million fatalities annually according to the World Health Organization [1]. These pathologies encompass a range of disorders affecting the heart and blood vessels, including coronary artery disease, cerebrovascular disease, and peripheral arterial disease, with atherosclerosis serving as the underlying pathological process in most cases.

The clinical manifestations often remain asymptomatic for decades, with acute events such as myocardial infarction or stroke frequently occurring as the first presentation of disease.

This silent progression underscores the critical importance of early risk identification and preventive intervention. Traditional risk stratification relies on established factors including age, hypertension, dyslipidaemia, smoking and diabetes mellitus. However, the complex interplay between these variables and individual patient characteristics creates challenges for accurate risk prediction using conventional statistical approaches alone [10].

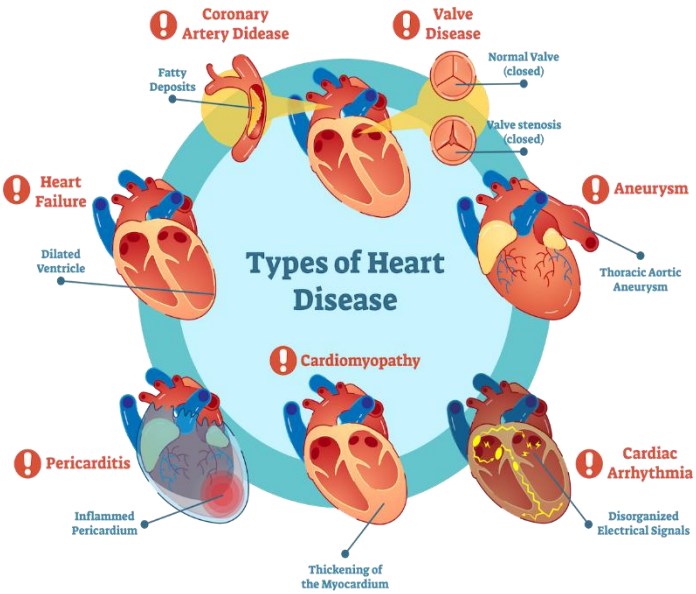


Figure 1: from https://www.udmi.net/cardiovascular-disease-risk/ - *Type of Heart Disease.* The image illustrates how various pathologies affect the heart's structure: ranging from valve restriction (stenosis) and arterial wall bulging (aneurysm) to inflammation of the outer tissue (pericarditis).

1.2 Machine learning applications in medical diagnostics

The integration of machine learning techniques into clinical decision-making has emerged as a transformative approach to improve diagnostic accuracy and risk stratification. Unlike traditional statistical methods that assume predefined relationships between variables, machine learning algorithms can identify complex, non-linear patterns within high-dimensional clinical data without explicit programming.

In the cardiovascular domain, predictive models have been successfully applied to interpret electrocardiographic signals, analyze medical imaging, and integrate electronic health record data for risk assessment. The ability of these algorithms to process multiple clinical parameters simultaneously and detect subtle interactions between risk factors offers potential advantages over conventional scoring systems [2].

1.3 Aim of the project

The primary objective of this study is to develop a machine learning pipeline capable of predicting cardiovascular disease from routine clinical data, with the ultimate goal of creating a practical tool to support early diagnosis in clinical settings. By systematically comparing multiple classification approaches and evaluating their performance against clinically meaningful metrics, this work seeks to identify the model that best balances high sensitivity—essential for capturing at-risk patients—with the specificity required to avoid unnecessary interventions. Ultimately, this framework aims to demonstrate how machine learning can augment clinical judgment, enabling earlier detection of heart disease and more timely preventive care, contributing to improved patient outcomes and reduced healthcare burden.

2. Material and Methods:

2.1 Visual pipeline

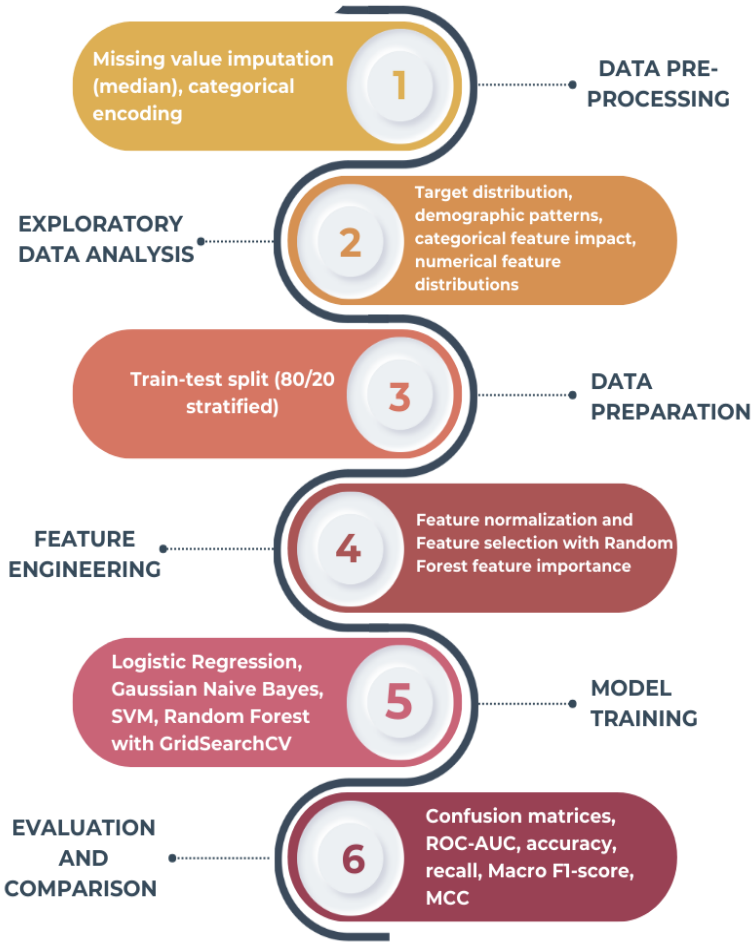


Figure 2: *Main stages of the Machine Learning pipeline*. The infographic outlines the end-to-end workflow for model development: from initial data pre-processing and exploratory analysis to feature engineering, model training and final evaluation.

2.2 Data Pre-processing

The dataset was obtained from the UCI Machine Learning Repository and consists of 920 patient records aggregated from four independent cohorts (Cleveland, Hungarian, Switzerland, and VA Long Beach). Each observation includes 13 clinical attributes and a target variable encoding heart disease severity on an ordinal scale from 0 (absence) to 4 (severe disease).

Missing values, originally encoded as "?", were addressed using median imputation applied independently to each feature. This approach preserves the full dataset while minimizing distortion from skewed distributions and extreme clinical values. Unlike mean imputation, the median provides a robust estimate of central tendency and reduces sensitivity to outliers, which are common in biomedical measurements.

Categorical variables originally encoded numerically were transformed into descriptive labels to improve interpretability and ensure consistency during exploratory analysis. This transformation did not alter the statistical properties of the data but facilitated clinically meaningful interpretation of model inputs.

2.3 Exploratory Data Analysis

EDA was conducted to characterize the dataset and identify relevant patterns prior to modeling, guiding feature selection, detecting potential data quality issues, and informing appropriate preprocessing strategies.

2.3.1 Target Variable Transformation

The original five-class target variable was transformed into a binary outcome representing the presence (classes 1–4) or absence (class 0) of heart disease. This choice reflects the study's focus on clinical screening rather than severity stratification, and addresses the sparsity inherent in higher severity categories. Following binarization, the dataset presents a mild class imbalance, with 44.7% healthy and 55.3% disease-positive cases, which was accounted for in all subsequent modeling and evaluation choices.

2.3.2 Analysis of Predictors

Both categorical and numerical variables were systematically explored to assess their relationship with the target variable.

- ***Categorical variables*** were analyzed through normalized contingency tables, allowing estimation of conditional probabilities of disease given each category. This approach provides an interpretable measure of association without assuming linear relationships.
- **Numerical variables** were examined using distributional summaries (median, interquartile range) and visual comparisons across classes. This enabled identification of shifts in central tendency and dispersion associated with disease status.
- **Demographic variables** (age and sex) were analyzed separately due to their established clinical relevance, with emphasis on identifying trends consistent with known epidemiological patterns.

Overall, this phase informed subsequent feature engineering decisions and highlighted variables likely to contribute to predictive performance.

2.4 Data Preparation

2.4.1 Outlier Detection and Treatment

Outliers in continuous variables were identified using the Z-score method, which standardizes each observation relative to the feature mean and standard deviation. Observations with $|Z| > 3$ were considered extreme and removed, as this threshold corresponds to the theoretical bounds of a normal distribution and limits the influence of atypical values on model training. Following this step, 12 records were excluded, yielding a final dataset of 908 observations

2.4.2 Encoding of Categorical Variables

Categorical features were transformed using one-hot encoding via `pd.get_dummies()`, generating binary indicator variables for each category. To avoid multicollinearity, one reference category per variable was omitted (`drop_first=True`). Following encoding, the feature space expanded from 13 original clinical attributes to 17 binary and continuous predictors, all represented numerically and suitable for machine learning algorithms.

2.4.3 Data Splitting

The 908-record dataset was partitioned into training (726 records, 80%) and test (182 records, 20%) subsets using stratified sampling, preserving the class distribution across both sets. In the training set, 327 patients were classified as healthy and 399 as disease-positive; in the test set, 82 were healthy and 100 disease-positive.

All preprocessing steps involving learned parameters were fitted exclusively on the training set and applied unchanged to the test set, preventing information leakage and ensuring unbiased performance estimates.

2.5 Feature Engineering

Following data preparation, feature engineering refined the input space to enhance model performance. Normalization was applied to equalize feature contributions across different scales, while feature selection reduced dimensionality by retaining only the most clinically and statistically informative variables.

2.5.1 Feature Scaling

Continuous variables were normalized using Min-Max scaling to the range [0, 1], applied to seven numerical features (age, resting blood pressure, cholesterol, maximum heart rate, ST depression, number of major vessels, and thalassemia type).

This transformation ensures proportional contribution during model training, particularly for algorithms based on distance metrics or gradient optimization. Scaling parameters were estimated on the training data and applied unchanged to the test set.

2.5.2 Feature Selection

Feature selection was performed using Random Forest feature importance as an embedded method. A Random Forest classifier with 100 trees was trained on all 17 features, and each predictor was ranked by its contribution to Mean Decrease in Impurity (Gini importance) across the ensemble.

The top 10 features by importance score were retained for model training: cholesterol, maximum heart rate, age, ST depression, exercise-induced angina, atypical angina chest pain, resting blood pressure, thalassemia type, sex, and non-anginal chest pain. Lower-ranked variables were excluded as they provided minimal additional predictive signal. This approach captures both linear and non-linear relationships while producing a compact, clinically interpretable feature set.

2.6 Model Training

Four classification algorithms were selected to span a broad range of modeling paradigms: Logistic Regression (linear) [3], Gaussian Naive Bayes (probabilistic) [6], Support Vector Machine (margin-based) [5], and Random Forest (ensemble tree-based) [4]. This diversity was intentional — by covering fundamentally different inductive biases, the comparison allows assessment of whether predictive performance is driven by data structure or by algorithmic complexity. All models were trained on the same 10-feature set and binary target variable to ensure comparability.

Hyperparameters were optimized using grid search with stratified 5-fold cross-validation, with ROC-AUC as the optimization criterion. Cross-validation reduces variance in performance estimates and ensures that hyperparameter selection is not dependent on a single data split.

2.7 Model Evaluation

All classifiers were evaluated on the held-out test set (182 records) to assess generalization performance. Final metrics were computed exclusively on this subset, independently of the cross-validation procedure used during training.

2.7.1 Evaluation Metrics

Performance was assessed using four complementary metrics.

Accuracy measures the overall proportion of correctly classified instances and is defined as:

$$\frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

While widely used, accuracy alone may be insufficient in medical contexts, as it does not distinguish between different types of classification errors.

Recall (Sensitivity) quantifies the model's ability to correctly identify patients with heart disease:

$$\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

This metric is particularly critical in clinical screening scenarios, where false negatives (missed diagnoses) carry significant risk.

Macro F1-Score represents the harmonic mean of precision and recall, computed independently for each class and then averaged:

$$\frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

The macro-averaging approach assigns equal importance to both classes, making it suitable for moderately imbalanced datasets.

Matthews Correlation Coefficient (MCC) provides a balanced evaluation by considering all elements of the confusion matrix:

$$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Unlike accuracy, MCC remains informative even when class distributions are uneven, making it particularly appropriate for medical classification tasks [7].

2.7.2 Error Analysis and Model Comparison

Confusion matrices were used to quantify classification outcomes with particular attention to false negatives. Among the four models, Random Forest recorded the fewest false negatives (7), followed by SVM (9), Logistic Regression (11), and Gaussian Naive Bayes (12), reflecting meaningful differences in clinical sensitivity despite comparable overall accuracy.

ROC curves assessed discriminative ability across varying decision thresholds. AUC scores were closely clustered across all classifiers — SVM (0.913), Logistic Regression (0.911), Gaussian Naive Bayes (0.910), and Random Forest (0.909) — indicating that all algorithms captured the underlying data patterns with equivalent effectiveness. Model selection was therefore guided primarily by recall and MCC rather than AUC.

2.7.3 Reproducibility and Model Persistence

All trained models, the MinMaxScaler, and the final feature list were serialized using the joblib library, ensuring full reproducibility of the experimental pipeline and facilitating future deployment in clinical decision support systems.

3.Results

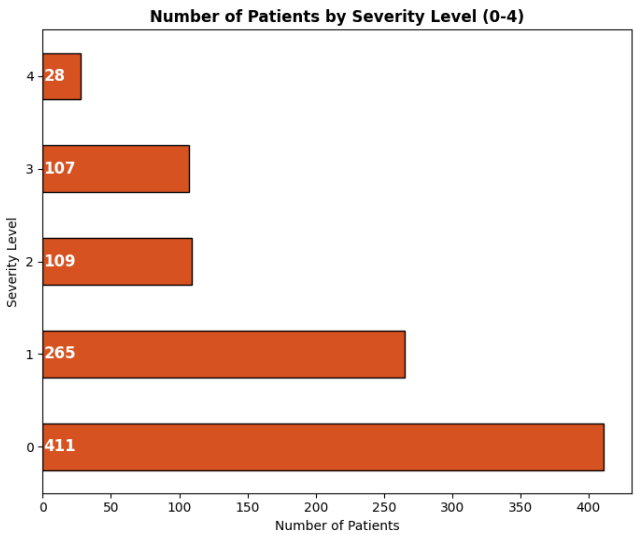
This section presents the results of the analytical workflow, encompassing exploratory data analysis, feature selection, model optimization, and performance evaluation. Findings are discussed with respect to their statistical validity and clinical interpretability in the context of cardiovascular disease (CVD) prediction.

3.1 Exploratory Data Analysis

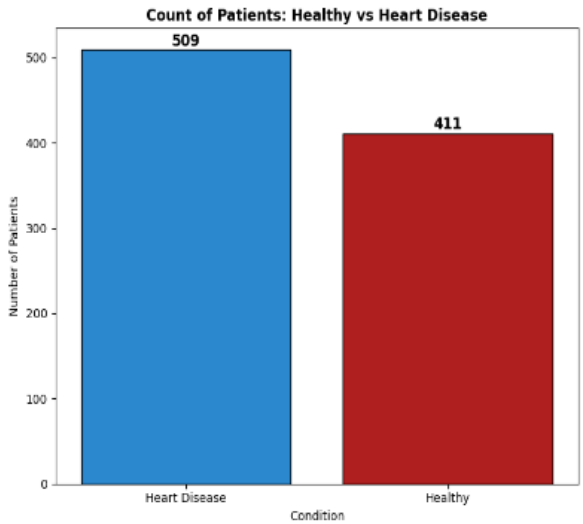
The analysis was conducted on the preprocessed dataset (n = 920), providing insights into the distribution of demographic and clinical variables and their association with disease presence.

3.1.1 Target Variable and Class Definition

The original target variable, defined on a five-level ordinal scale (0–4), exhibited a markedly imbalanced distribution, with higher severity levels progressively underrepresented. Healthy patients (level 0) accounted for 44.7% of the cohort, while mild disease (level 1) represented the largest disease subgroup at 28.8%. Moderate disease (levels 2–3) accounted for 23.4%, and severe cases (level 4) for the remaining 3.0% (n = 28).



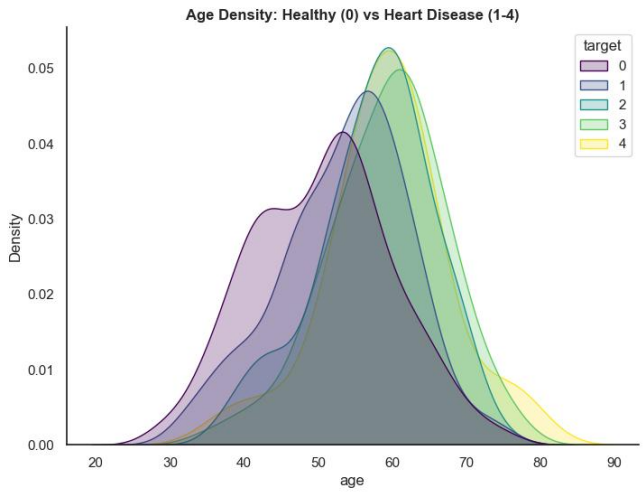
To address class sparsity and align with the screening objective of this study, the target was reformulated as a binary variable distinguishing healthy (0) from disease-positive (1–4) individuals. This transformation yielded a near-balanced distribution (55.3% disease vs. 44.7% healthy), improving statistical robustness, reducing potential majority-class bias, and enhancing clinical applicability by reflecting real-world binary triage scenarios.



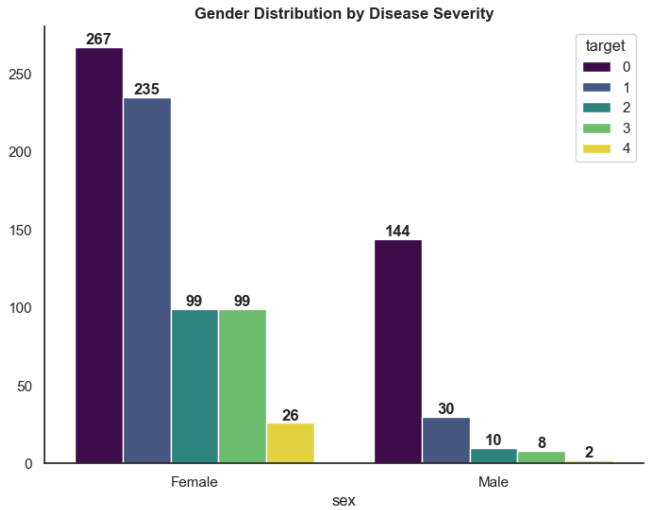
3.1.2 Demographic Factors

Age and sex distributions were consistent with established epidemiological patterns.

Age demonstrated a monotonic association with disease severity. Healthy individuals were predominantly concentrated between 45–65 years (peak: 55–60 years), while mild cases shifted toward 60–65 years, moderate cases toward 65–70 years, and severe cases (levels 3–4) toward patients older than 70. This progressive rightward shift across severity levels is consistent with established evidence that both prevalence and severity of cardiovascular disease increase with age.



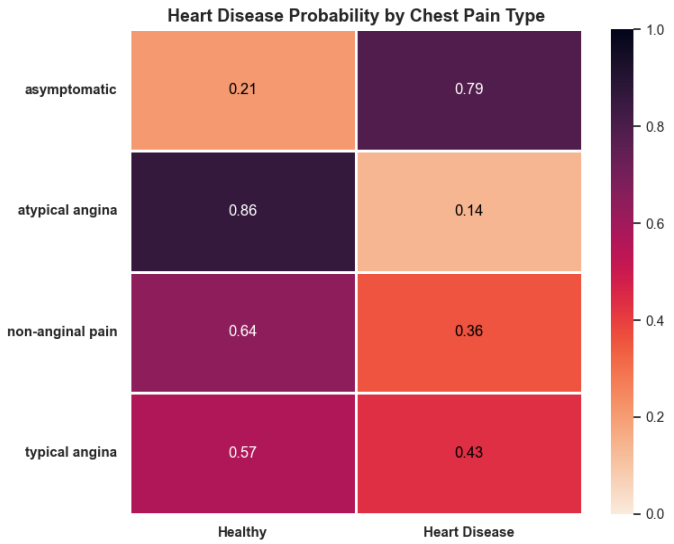
Sex-based analysis revealed a pronounced imbalance in both cohort composition and disease burden. Males comprised 71% of the total sample (n ≈ 653) and exhibited substantially higher disease prevalence: 78% of male patients were disease-positive, compared to only 12% of females. Severe disease (levels 3–4) affected 15.3% of male patients versus 5.2% of females, a nearly threefold difference. These patterns are consistent with known biological and hormonal factors — including protective estrogenic effects in premenopausal women — that modulate cardiovascular risk across sexes.



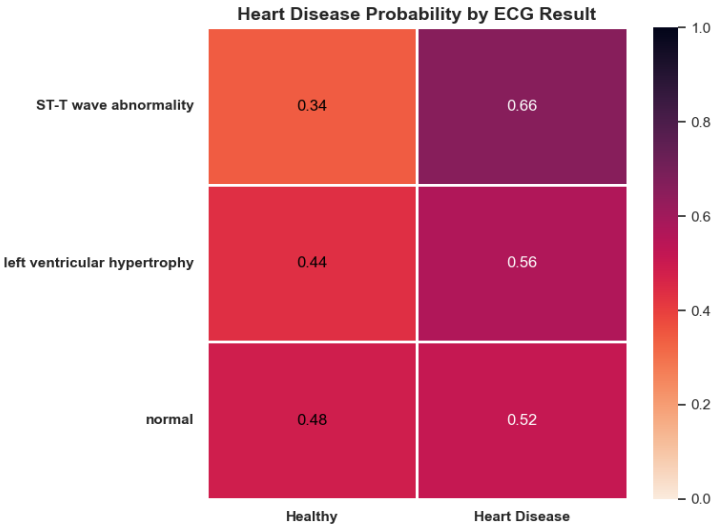
3.1.3 Categorical Variables

Categorical predictors showed heterogeneous discriminative power across disease categories.

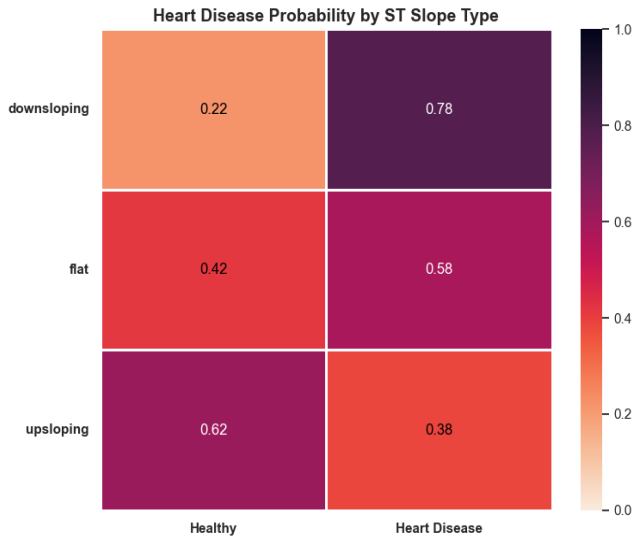
Chest pain type emerged as a clinically informative indicator with a counterintuitive pattern. Asymptomatic patients exhibited the highest disease probability (79%), consistent with the phenomenon of silent ischemia, in which significant coronary disease can be present without symptomatic expression. Typical angina showed only moderate association (43%), suggesting that classic symptoms do not reliably confirm disease in this cohort. Atypical angina demonstrated the weakest association (14%), likely reflecting non-cardiac etiologies such as musculoskeletal, gastrointestinal, or anxiety-related presentations.



Resting ECG features showed limited standalone discriminative power, with all three categories exhibiting disease probabilities exceeding 50%. ST-T wave abnormalities showed the strongest association (66%), while left ventricular hypertrophy (LVH) reached 56%, reflecting structural adaptation to chronic pressure overload. Notably, even patients with a normal ECG showed a 52% disease probability, underscoring the limited sensitivity of resting electrocardiography as a standalone screening tool.



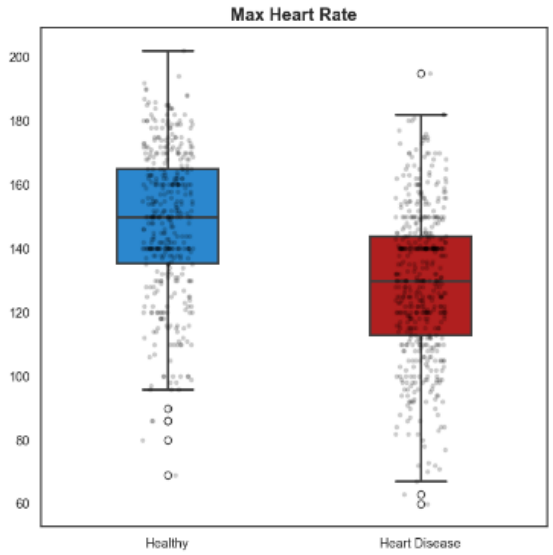
The ST segment slope displayed the clearest risk gradient among categorical features. Downsloping segments carried the highest disease probability (78%), consistent with their established association with exercise-induced ischemia. Flat segments showed a moderate association (58%), while upsloping segments — generally considered a normal exercise response — showed the lowest probability (38%).



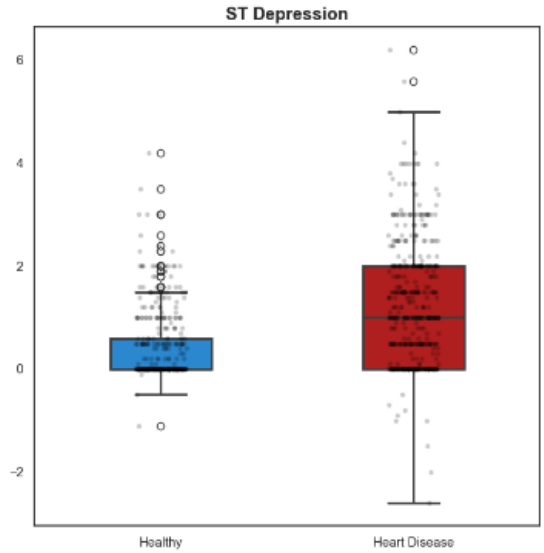
3.1.4 Numerical Variables

Among continuous variables, exercise-related features provided the strongest class separation.

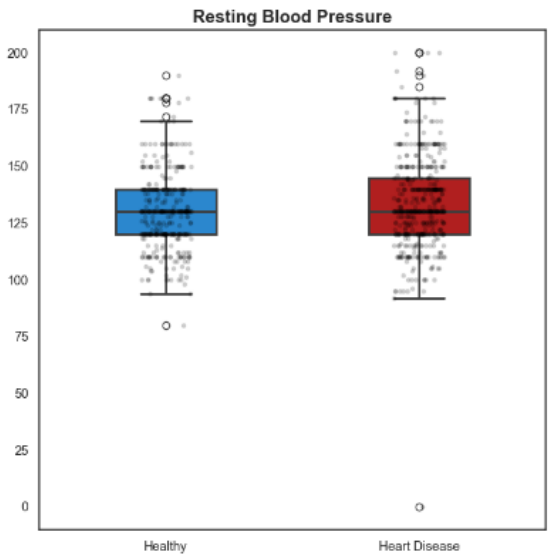
Maximum heart rate (`thalach`) was markedly lower in disease-positive patients, reflecting impaired chronotropic response and reduced cardiovascular fitness. Healthy patients exhibited higher median values and greater distributional spread, whereas disease patients showed compressed, lower-range values with occasional outliers. This pattern is consistent with the blunted cardiac output response characteristic of compromised myocardial function.



ST depression (`oldpeak`) provided the clearest binary separation of any continuous variable. Disease-positive patients concentrated at elevated values, reflecting exercise-induced ischemia, while healthy patients clustered near zero with minimal variation. Extreme outliers in the disease group likely correspond to severe or multi-vessel coronary involvement.



Resting blood pressure (`trestbps`) showed limited differentiation between classes, with overlapping medians and interquartile ranges in both groups. This finding suggests that resting blood pressure, while a recognized long-term cardiovascular risk factor, has limited discriminative utility as a standalone acute indicator — likely due to the episodic nature of hypertension and the confounding effect of antihypertensive therapy in treated patients.

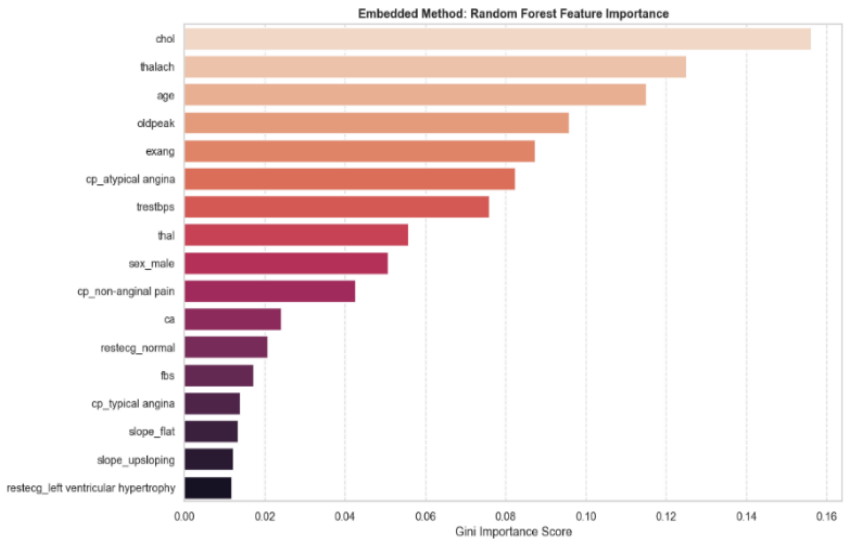


3.2 Feature Selection

Feature relevance was assessed using Random Forest feature importance as an embedded selection method. A Random Forest classifier with 100 trees was trained on all 17 features following one-hot encoding, and each predictor was ranked by its contribution to Mean Decrease in Impurity (Gini importance) across the ensemble.

The resulting importance hierarchy identified three tiers of predictive contribution. The top tier — cholesterol (0.156), maximum heart rate (0.125), and age (0.115) — collectively captured metabolic burden, functional cardiac capacity, and cumulative exposure risk. The second tier comprised exercise-related indicators: ST depression (0.096), exercise-induced angina (0.087), and atypical chest pain presentation (0.082). The third tier included moderate contributors such as resting blood pressure (0.076), thalassemia type (0.056), sex (0.051), and non-anginal chest pain (0.043). Features below this threshold — including the number of fluoroscopy-colored vessels, resting ECG, fasting blood sugar, and ST slope — showed limited predictive signal and were excluded.

The final feature set comprised the top 10 predictors: cholesterol, maximum heart rate, age, ST depression, exercise-induced angina, atypical chest pain, resting blood pressure, thalassemia type, sex, and non-anginal chest pain. These variables collectively capture three complementary dimensions of cardiovascular risk: metabolic dysregulation, functional stress response, and baseline clinical and demographic profile.



3.3 Model Training and Hyperparameter Optimization

Four classification models were trained on the selected feature set using the 726-record training subset. Hyperparameters were optimized via grid search with stratified 5-fold cross-validation, using ROC-AUC as the objective metric.

Logistic Regression converged to `C = 10`, reflecting relatively weak regularization and a flexible linear boundary. Gaussian Naive Bayes selected `var_smoothing = 1e-9`, indicating that variance estimates were already stable and required minimal smoothing. SVM was optimized at `C = 1` with an RBF kernel, a relatively strong regularization. Random Forest selected `max_depth = 5, min_samples_split = 5, and n_estimators = 100`, reflecting constrained tree complexity designed to limit overfitting while preserving non-linear interaction capture.

3.4 Model Evaluation

All models were evaluated on the independent test set (n = 182: 82 healthy, 100 disease-positive) using five complementary metrics: Accuracy, Recall, Macro F1-Score, Matthews Correlation Coefficient (MCC), and AUC-ROC [9].

3.4.1 Comparative Performance

All four models achieved comparable overall accuracy (84.6–85.2%), indicating that the dataset contains consistent and learnable predictive structure across fundamentally different algorithmic paradigms. AUC-ROC scores were tightly clustered in the range 0.909–0.913, confirming equivalent discriminative ability across decision thresholds.

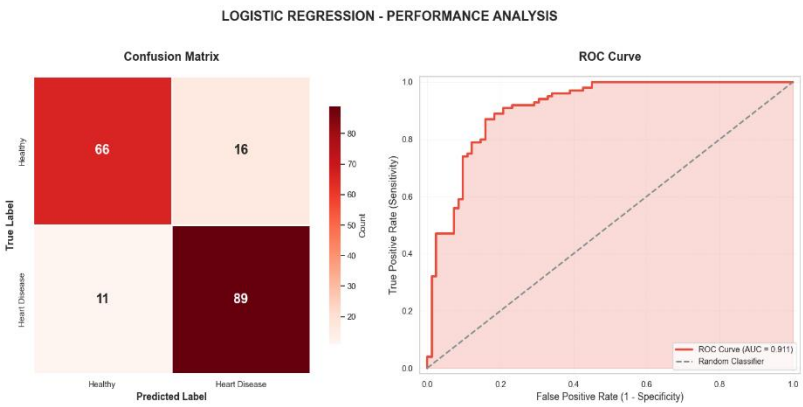
Meaningful differences emerged, however, in sensitivity-specificity trade-offs. Random Forest achieved the highest recall (0.930) and MCC (0.704), indicating superior ability to identify disease-positive patients while maintaining balanced overall prediction quality. SVM reached the highest AUC-ROC (0.913) but showed a lower MCC (0.690), reflecting a sensitivity-oriented decision boundary with increased false positive rate. Logistic Regression and Gaussian Naïve Bayes produced near-identical results across all metrics, confirming that a substantial portion of the predictive structure is linearly separable.

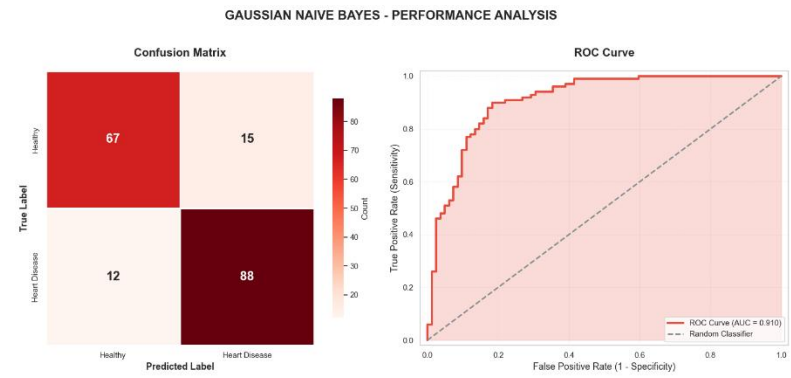
Model	Accuracy	Recall	Macro F1	MCC	AUC-ROC
Random Forest	0.852	0.930	0.847	0.704	0.909
Logistic Regression	0.852	0.890	0.849	0.700	0.911
Gaussian Naïve Bayes	0.852	0.880	0.850	0.700	0.910
SVM	0.846	0.910	0.842	0.690	0.913

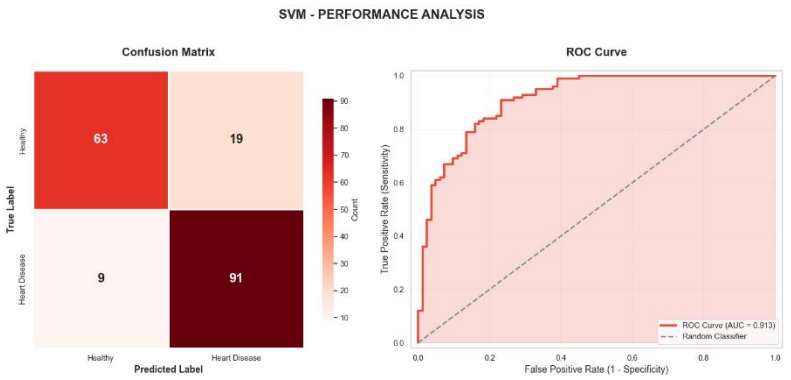
3.4.2 Error Analysis

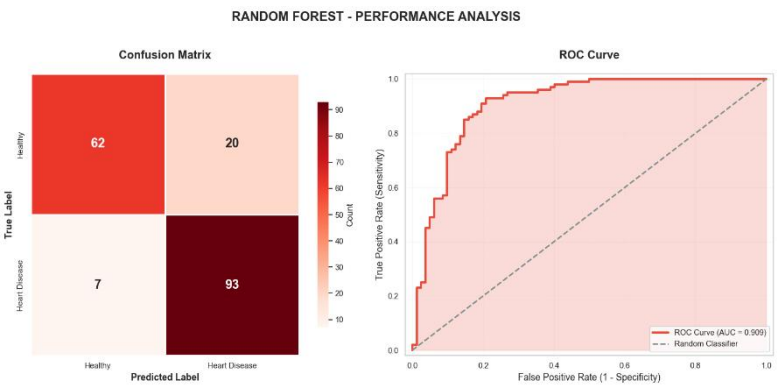
Confusion matrix analysis revealed that Random Forest minimized false negatives (n = 7), the most clinically critical error type in screening contexts as they correspond to missed diagnoses. SVM produced 9 false negatives, while Logistic Regression and Gaussian Naïve Bayes recorded 11 and 12 respectively. The latter two models achieved a more balanced error distribution across false positive and false negative rates, while Random Forest and SVM traded higher false positives for improved disease detection.

ROC curve analysis confirmed strong discriminative performance for all classifiers, with all curves rising steeply toward the top-left region and no single model dominating across all thresholds. The negligible AUC differences suggest that the dataset's predictive signal is robust and consistently captured regardless of the modeling paradigm employed [8].









3.4.3 Model Selection

Based on its superior recall, highest MCC, and minimal false negative rate, **Random Forest** was selected as the final model for cardiovascular risk prediction. In clinical screening applications, where the cost of a missed diagnosis substantially outweighs that of a false positive referral, these properties represent the decisive performance advantage over competing models of otherwise comparable accuracy. All trained models, preprocessing components, and selected feature sets were serialized using the `joblib` library to ensure full reproducibility and facilitate future deployment within clinical decision support systems.

4. Discussion

This study developed and evaluated a complete machine learning pipeline for binary cardiovascular disease classification using clinical data from 908 patients. The findings demonstrate that reliable predictive performance is achievable with routine clinical measurements, and that algorithmic complexity does not necessarily translate into superior diagnostic utility.

4.1 Model Performance and Clinical Implications

All four classifiers achieved comparable accuracy (84.6–85.2%) and AUC-ROC scores within a narrow range (0.909–0.913), indicating that the dataset contains consistent, learnable structure accessible to models of varying complexity. This convergence suggests that the selected feature set captures the underlying biological signal effectively, regardless of the inductive bias of the algorithm applied.

Despite equivalent overall accuracy, clinically meaningful differences emerged in sensitivity-specificity trade-offs. Random Forest was selected as the final model primarily on the basis of its superior recall (0.930) and MCC (0.704), and its minimal false negative count (n = 7). In screening contexts, false negatives — patients with disease classified as healthy — carry substantially greater clinical risk than false positives, as they result in missed diagnoses and delayed intervention. Under this criterion, the marginal performance advantage of Random Forest over simpler alternatives is clinically significant, even if statistically modest.

The competitive performance of Logistic Regression and Gaussian Naïve Bayes (both achieving 85.2% accuracy and MCC = 0.700) is noteworthy. It suggests that a substantial portion of the predictive structure in this dataset is linearly separable, and that model interpretability need not be sacrificed for performance. In resource-constrained clinical environments, a well-calibrated logistic regression model may offer a pragmatic and transparent alternative to ensemble methods.

4.2 Feature Importance and Clinical Interpretability

The three dominant predictors identified — cholesterol, maximum heart rate, and age — align well with established cardiovascular risk frameworks. Elevated cholesterol reflects atherogenic burden and plaque accumulation; reduced maximum heart rate indicates impaired chronotropic reserve and diminished myocardial perfusion under stress; age captures cumulative biological exposure and vascular aging. Their collective dominance over other features validates the physiological grounding of the model.

The strong predictive contribution of exercise-related variables (ST depression, exercise-induced angina) confirms that functional stress-testing responses are more informative than resting measurements alone. This is consistent with current cardiology practice, in which exercise tolerance testing remains a cornerstone of non-invasive ischemia evaluation. Conversely, resting blood pressure showed limited discriminative utility, likely reflecting the confounding effects of antihypertensive therapy and the episodic nature of hypertension.

The counterintuitive finding that asymptomatic chest pain presentation was associated with the highest disease probability (79%) highlights the phenomenon of silent ischemia, in which significant coronary disease develops without symptomatic expression. This finding has direct clinical relevance: it underscores the inadequacy of symptom-based screening alone and supports the value of objective functional and biochemical assessment in at-risk populations.

4.3 Limitations

Several limitations of this study warrant consideration. The dataset aggregates records from four geographically and temporally distinct cohorts (Cleveland, Hungarian, Switzerland, VA Long Beach), introducing potential heterogeneity in measurement protocols, patient demographics, and standard-of-care practices. Generalizability to contemporary clinical populations may therefore be limited.

The substantial proportion of missing values — addressed through median imputation — represents a further limitation. While this approach preserves dataset size and is robust to outliers, it may introduce systematic bias if missingness is non-random, a common pattern in clinical data where absence of a measurement is often clinically informative.

The binary reformulation of the target variable, while appropriate for screening purposes, discards prognostic information about disease severity. Future work incorporating ordinal or multi-class classification frameworks could provide more granular risk stratification. Additionally, the class imbalance between severity levels (particularly the underrepresentation of level 4, n = 28) limits the model's ability to characterize severe disease patterns.

Finally, the absence of external validation on an independent dataset limits assessment of true generalization performance. Cross-validated results are reported on the same source population, which may not fully capture variability in real-world deployment settings.

5. Conclusion

This study demonstrated that machine learning models trained on routine clinical measurements can achieve strong and reliable cardiovascular disease classification performance. Across four algorithmic paradigms — linear, probabilistic, margin-based, and ensemble — accuracy ranged from 84.6% to 85.2% and AUC-ROC from 0.909 to 0.913, confirming that the dataset's predictive structure is robust and consistently learnable.

Random Forest was selected as the optimal model for clinical deployment, on the basis of its superior sensitivity (recall = 0.930), highest MCC (0.704), and minimal false negative count — properties of primary importance in screening applications where missed diagnoses carry the greatest clinical cost. The feature importance analysis identified cholesterol, maximum heart rate, and age as the dominant predictors, with exercise-induced variables collectively constituting a second critical tier. These findings are consistent with established cardiovascular risk evidence and reinforce the diagnostic value of functional stress-testing parameters relative to resting measurements.

The competitive performance of simpler models suggests that interpretability and predictive accuracy are not inherently in tension for this classification task, a relevant consideration for clinical adoption. The full pipeline — including preprocessing, feature selection, model training, and serialization — is fully reproducible and designed for integration into clinical decision support systems.

Future work should address the identified limitations through external validation on independent, prospective cohorts; exploration of ordinal classification frameworks to preserve

severity information; and investigation of more sophisticated imputation strategies to handle clinically informative missingness. Integration of additional biomarkers — such as troponin levels, imaging-derived indices, or genetic risk scores — could further enhance predictive performance and clinical applicability.

6. References

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